



Heart Attack

"Model Comparison" Results

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Model Comparison

Champion	Name	Algorithm Name	Accuracy
true	Ensemble	Ensemble	0.8858
false	Gradient Boosting - Bill (Hard)	Gradient Boosting	0.8857
false	Bayesian Network - Bill (Simple)	Bayesian Network	0.8856

Average Squared Error	Area Under ROC	Cumulative Lift	Cumulative Captured Response Percentage
0.0835	0.9544	1.9948	19.9478
0.0835	0.9544	1.9950	19.9502
0.0847	0.9534	1.9938	19.9384

Cutoff	Data Role	Depth	F1 Score
0.5000	VALIDATE	10	0.8837
0.5000	VALIDATE	10	0.8836
0.5000	VALIDATE	10	0.8836

False Discovery Rate	False Positive Rate	Gain	Gini Coefficient
0.0999	0.0963	0.9948	0.9088
0.0994	0.0957	0.9950	0.9087
0.1010	0.0976	0.9938	0.9068

ROC Separation	KS (Youden)	Lift	Misclassification Rate
0.7716	0.7733	1.9921	0.1142
0.7715	0.7727	1.9932	0.1143
0.7711	0.7720	1.9890	0.1144

Multi-Class Log Loss	Misclassification at Cutoff	Misclassification Rate (Event)	Number of Observations
0.2753	0.1142	0.1142	169,472

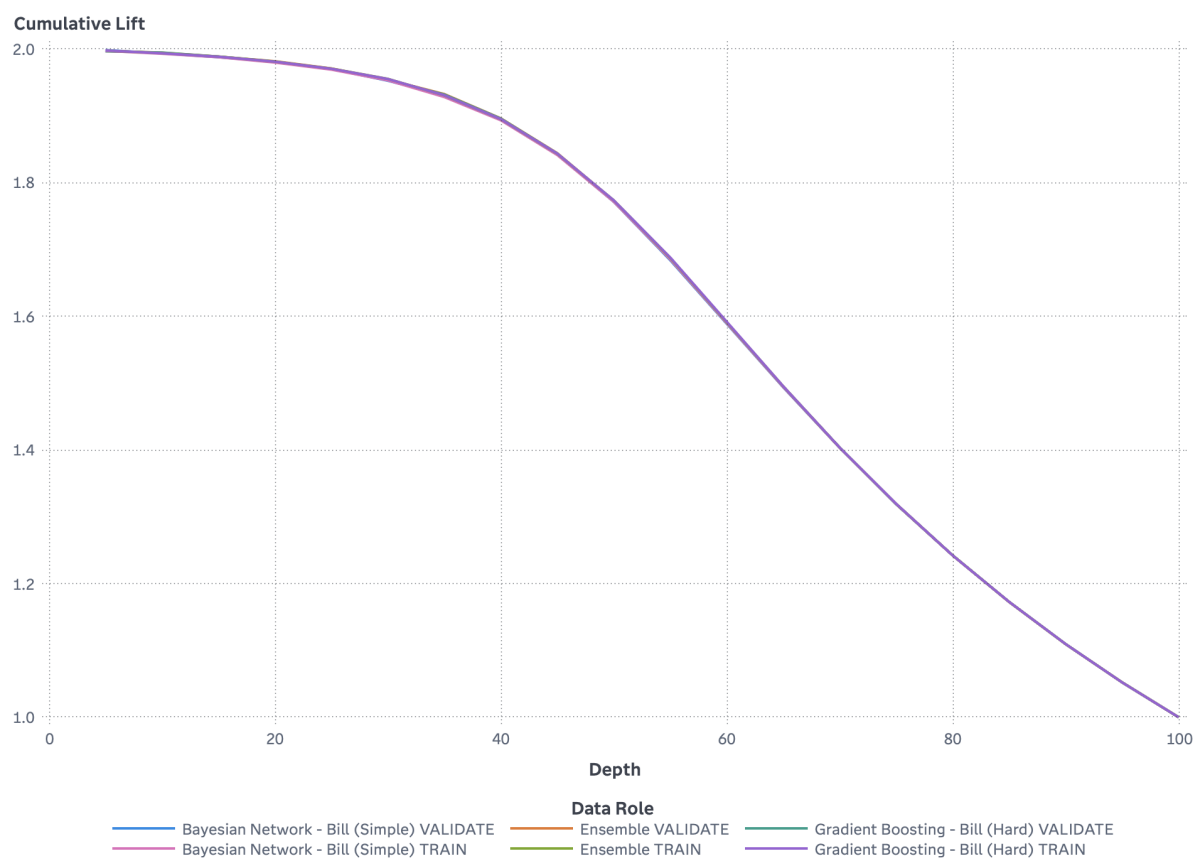
Multi-Class Log Loss	Misclassification at Cutoff	Misclassification Rate (Event)	Number of Observations
0.2747	0.1143	0.1143	169,472
0.2831	0.1144	0.1144	169,472

Root Average Squared Error	Captured Response Percentage
0.2890	9.9603
0.2889	9.9662
0.2910	9.9450

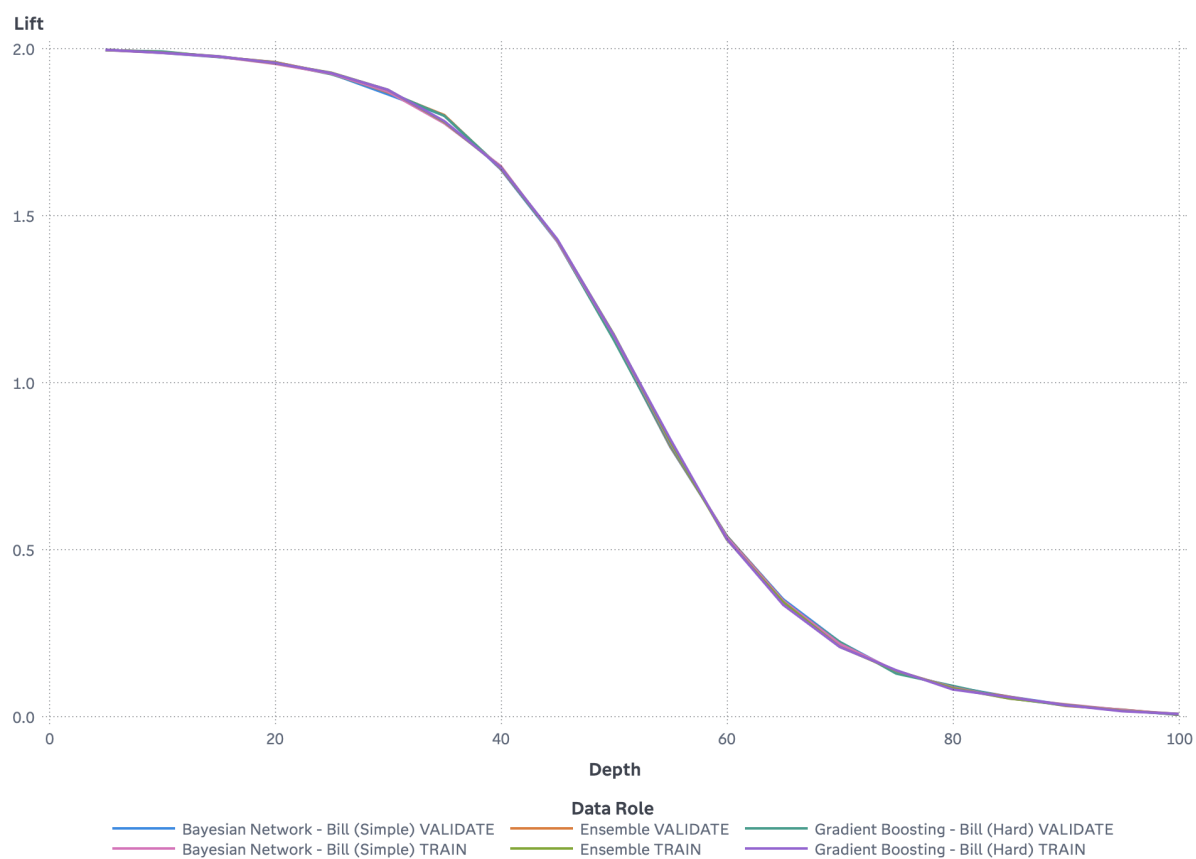
Properties

Property Name	Property Value
selectionCriteriaClass	Accuracy
selectionCriteriaInterval	Average squared error
selectionTable	Validate
selectionDepth	10
cutoff	0.50

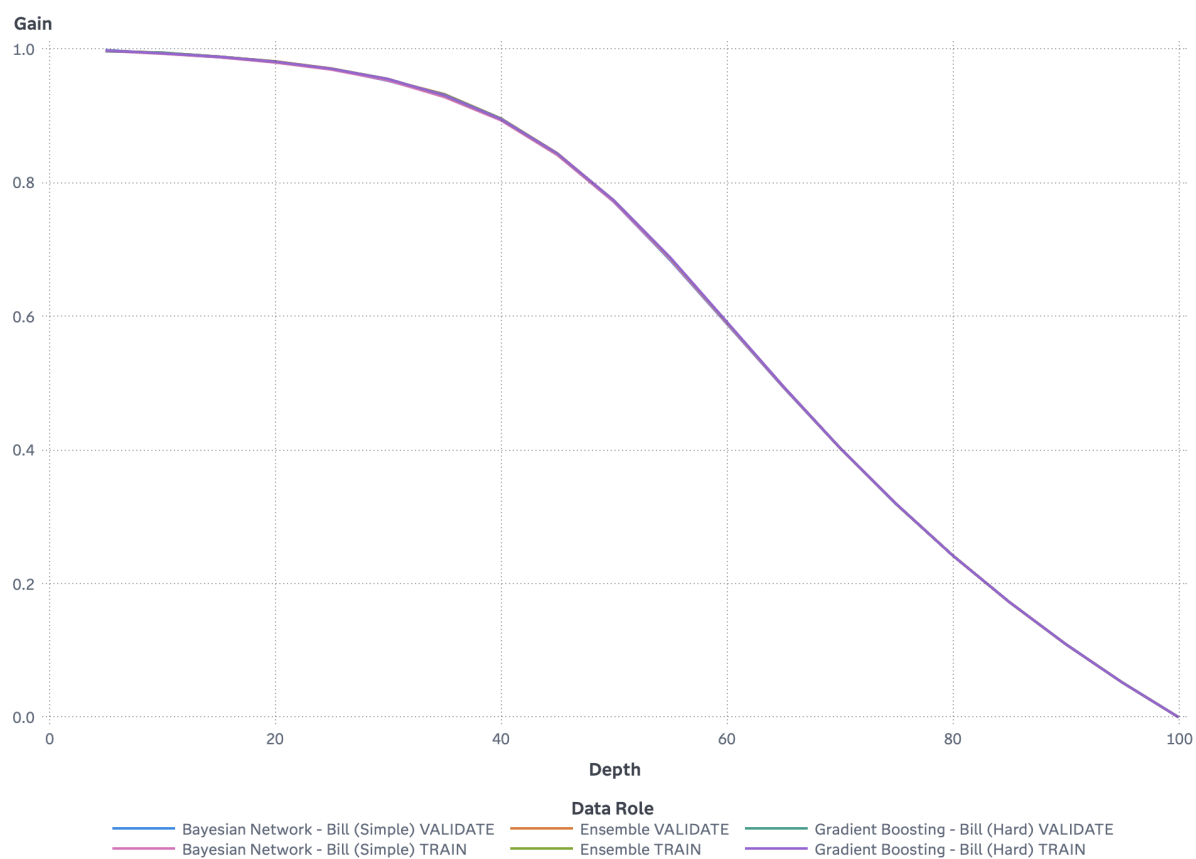
Cumulative Lift



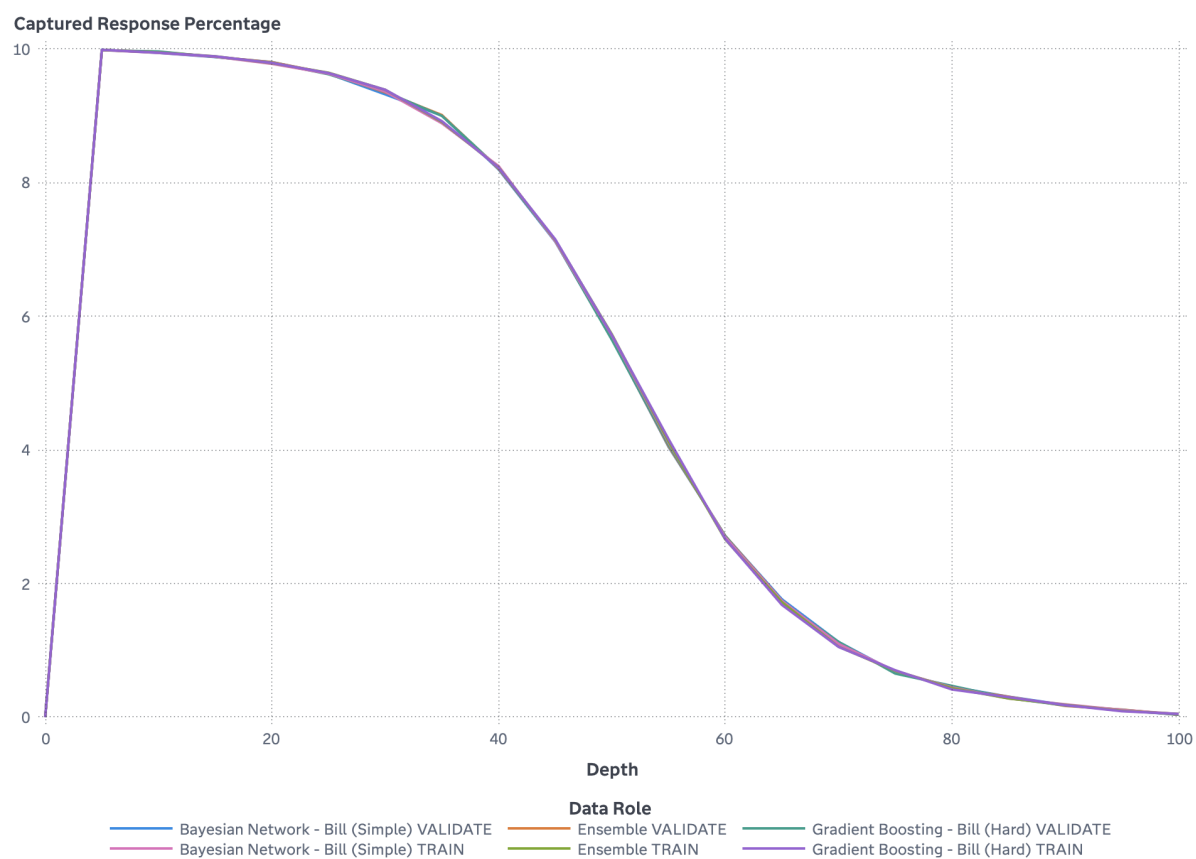
Lift



Gain

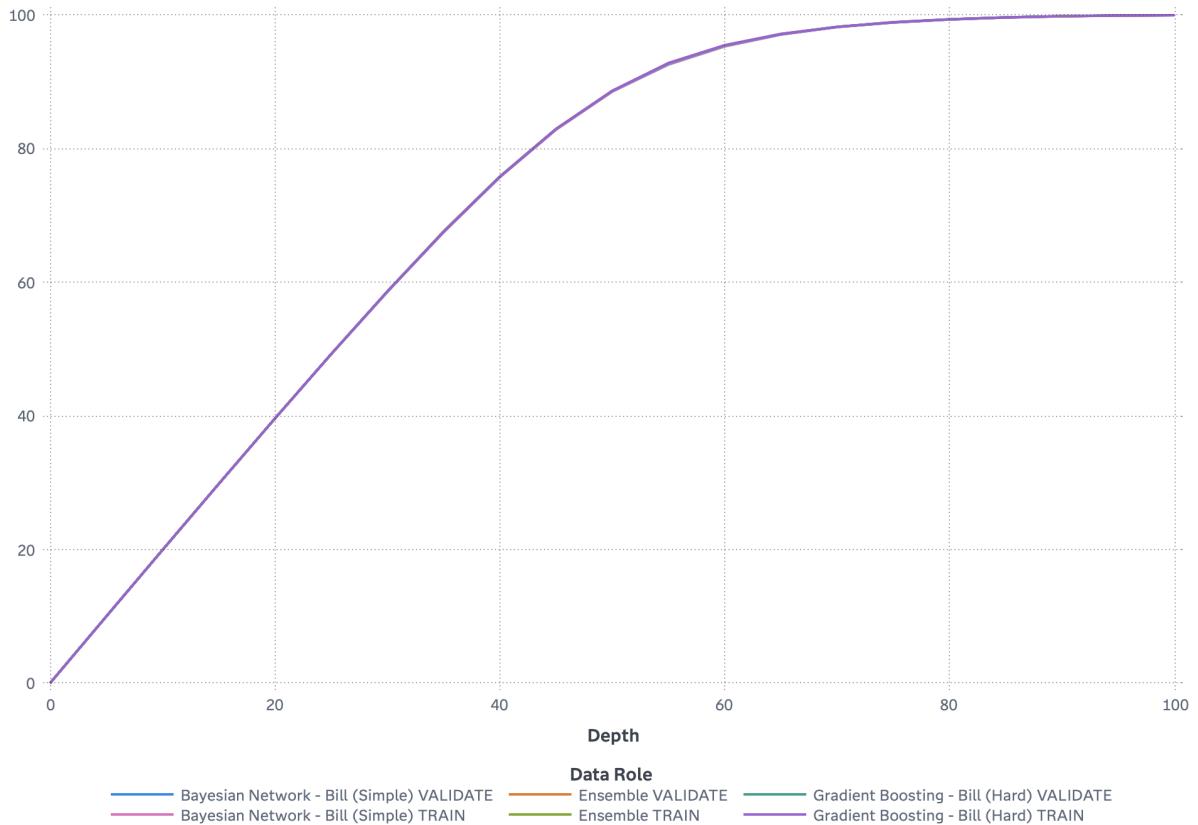


Captured Response Percentage

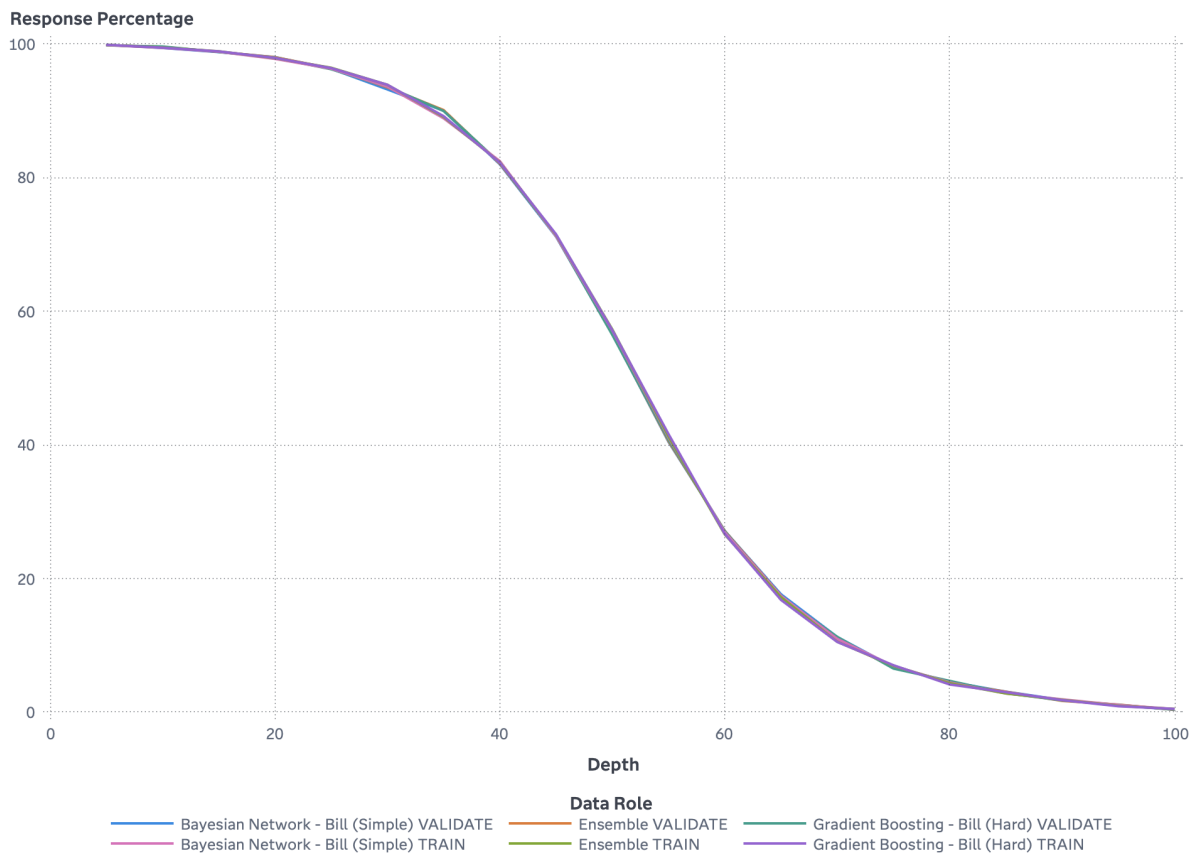


Cumulative Captured Response Percentage

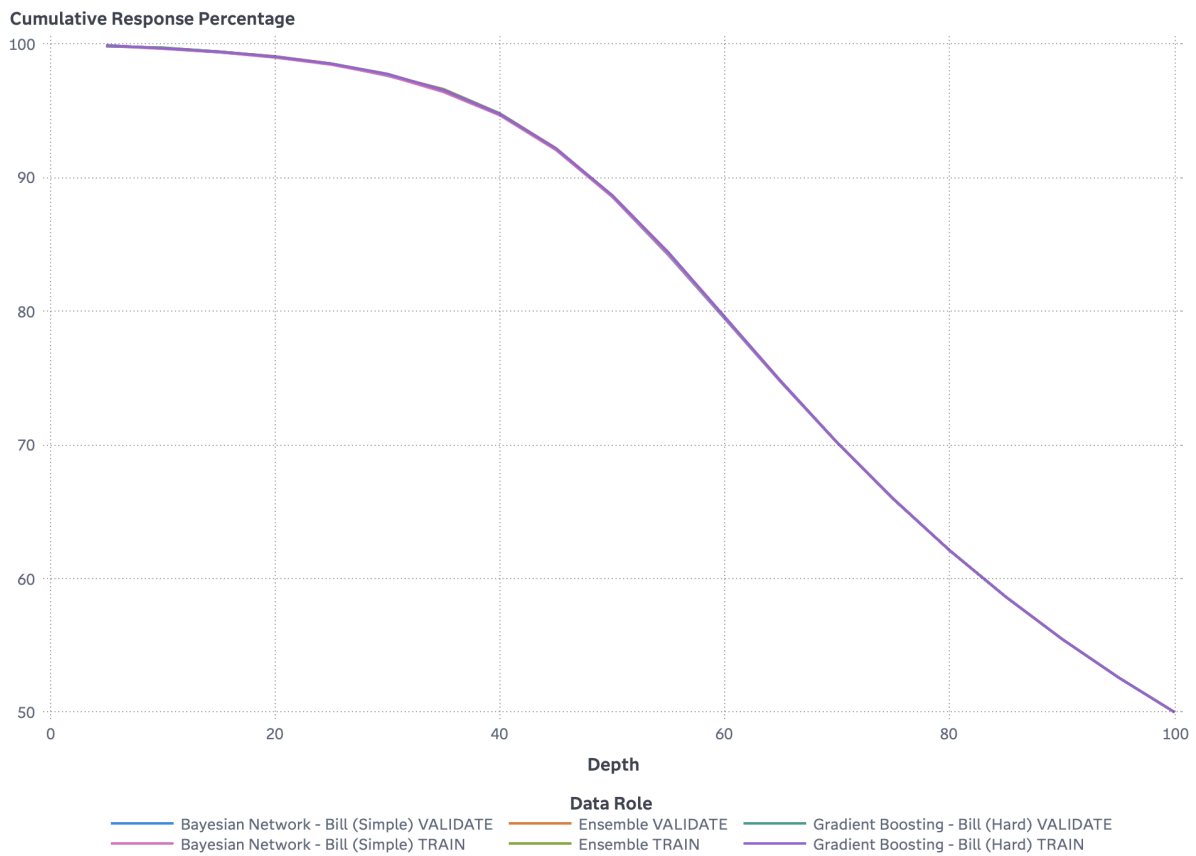
Cumulative Captured Response Percentage



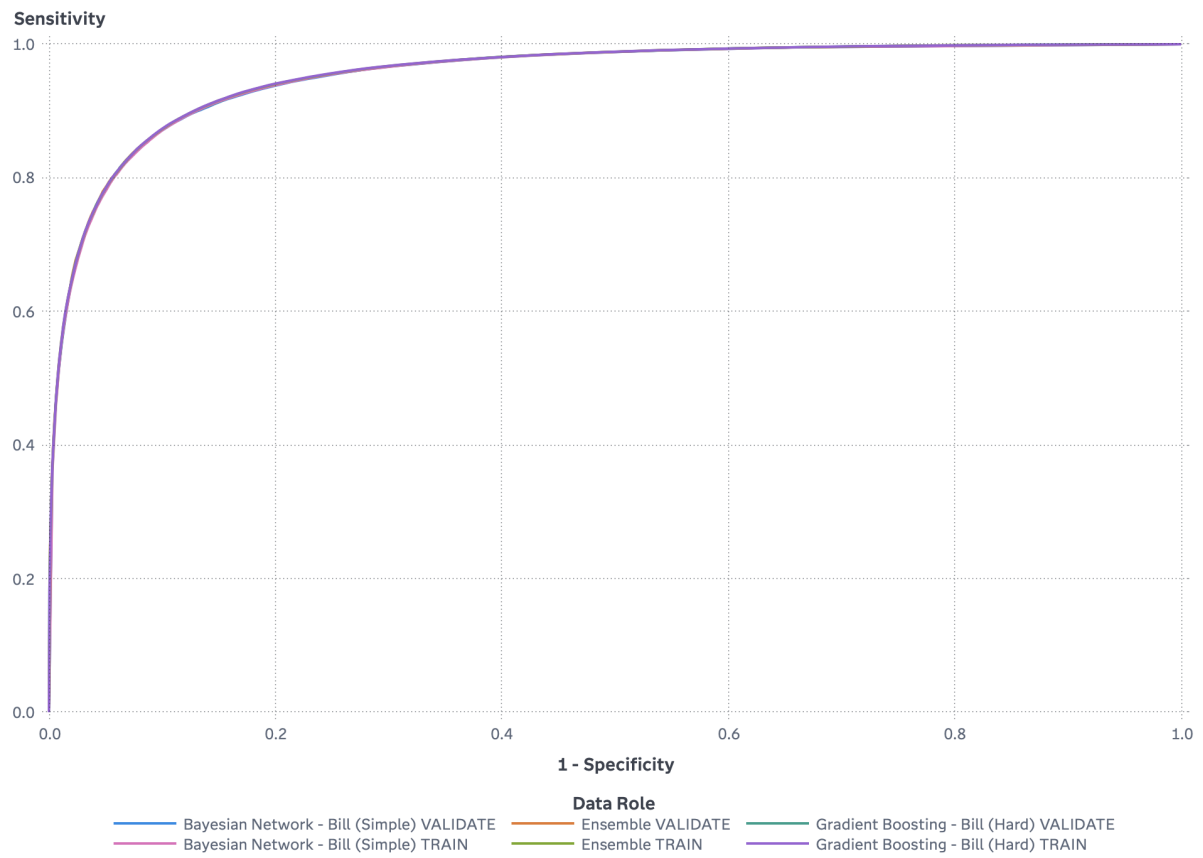
Response Percentage



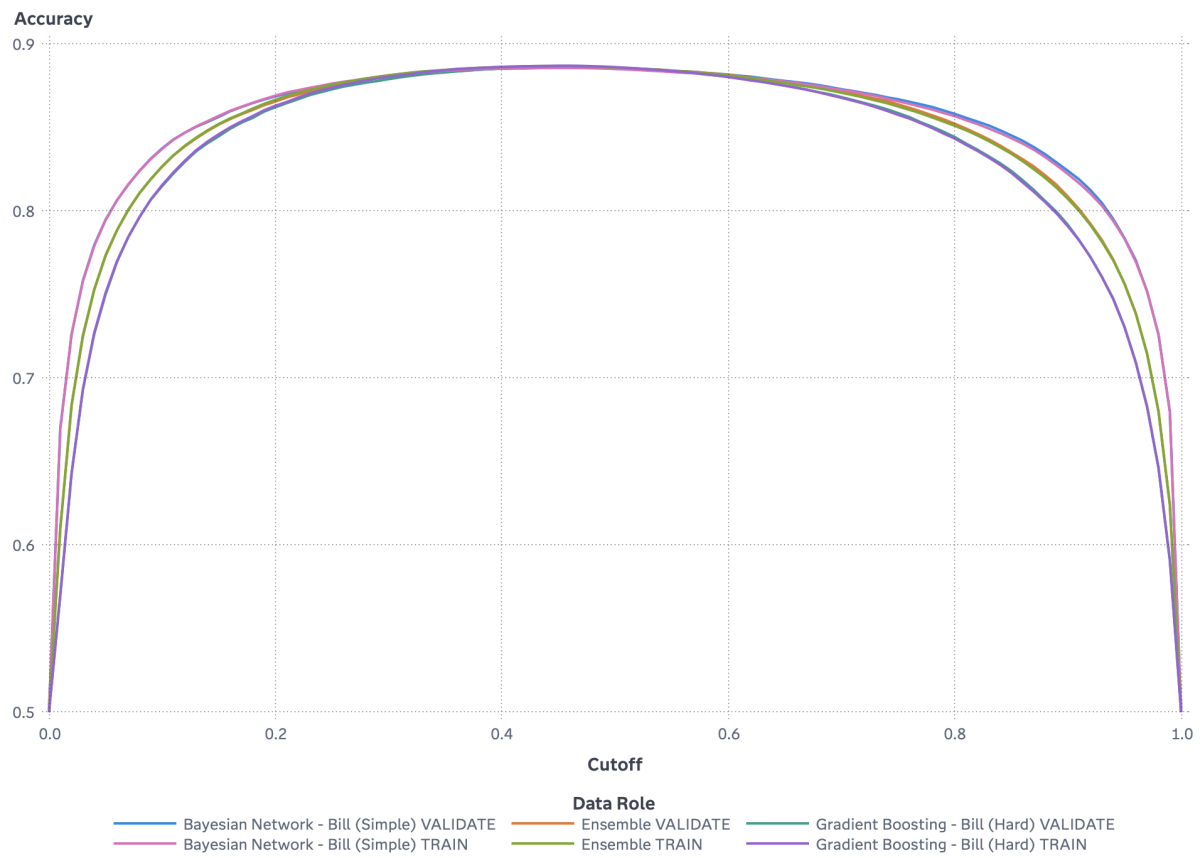
Cumulative Response Percentage



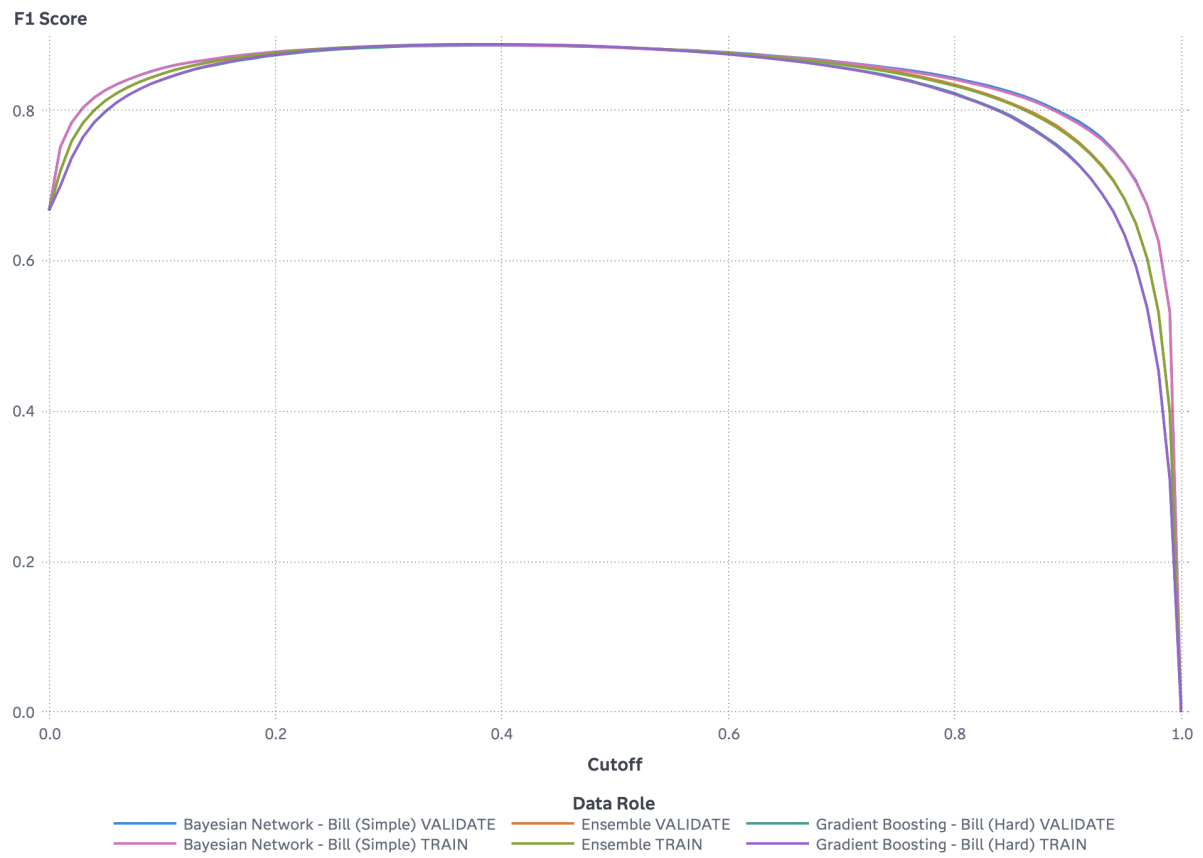
ROC



Accuracy



F1 Score



Fit Statistics

Statistics Label	TRAIN_Bayesian_Network - Bill (Simple)	VALIDATE_Bayesian_Network - Bill (Simple)	TRAIN_Gradient_Boosting - Bill (Hard)
Area Under ROC	0.9533	0.9534	0.9548
Average Squared Error	0.0847	0.0847	0.0830
Divisor for ASE	395,436	169,472	395,436
Formatted Partition	1	0	1
Gamma	0.9118	0.9121	0.9127
Gini Coefficient	0.9065	0.9068	0.9096
KS (Youden)	0.7717	0.7720	0.7743
KS at User-Specified Cutoff	0.7703	0.7711	0.7726
KS Cutoff	0.4500	0.4400	0.4600
Misclassification Rate	0.1149	0.1144	0.1137
Misclassification Rate (Event)	0.1149	0.1144	0.1137
Misclassification Rate at KS Cutoff (Event)	0.1141	0.1140	0.1128
Multi-Class Log Loss	0.2836	0.2831	0.2733
Number of Observations	395,436	169,472	395,436
Partition Indicator	1	0	1
Root Average Squared Error	0.2911	0.2910	0.2881
Target Name	Heart Disease	Heart Disease	Heart Disease
Tau	0.4533	0.4534	0.4548

VALIDATE_Gradient_Boosting - Bill (Hard)	TRAIN_Ensemble	VALIDATE_Ensemble
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VALIDATE_Gradient_Boosting - Bill (Hard)	TRAIN_Ensemble	VALIDATE_Ensemble
0.9544	0.9546	0.9544
0.0835	0.0833	0.0835
169,472	395,436	169,472
0	1	0
0.9118	0.9128	0.9124
0.9087	0.9092	0.9088
0.7727	0.7739	0.7733
0.7715	0.7721	0.7716
0.4400	0.4400	0.4400
0.1143	0.1140	0.1142
0.1143	0.1140	0.1142
0.1136	0.1130	0.1134
0.2747	0.2747	0.2753
169,472	395,436	169,472
0	1	0
0.2889	0.2886	0.2890
Heart Disease	Heart Disease	Heart Disease
0.4544	0.4546	0.4544